



Special Notice
Information Innovation Office
Request for Information

RFI: Levels of Resilience through Formal Methods

DARPA-SN-26-71

May 04, 2026

REQUEST FOR INFORMATION (RFI) DESCRIPTION

The Defense Advanced Research Projects Agency (DARPA) Information Innovation Office (I2O) seeks information on approaches, tools, and practices for achieving graduated levels of software resilience and assurance aligned to mission needs, cost, and lifecycle constraints. Respondents should provide information, technical approaches, analyses, and recommendations describing how graduated, incrementally adoptable levels of increasingly rigorous, mechanized, and automated software assurance can be defined, implemented, measured, and sustained across development, sustainment, and modernization activities. Responses should clearly articulate practical implementation strategies, measurable outcomes, and transition considerations associated with each proposed assurance level.

RFI BACKGROUND

Modern software systems continue to exhibit pervasive vulnerabilities, including those that persist despite extensive testing and compliance with conventional best practices. These vulnerabilities often arise from emergent behaviors, incomplete understanding of software structure and interfaces, unsafe data handling, and memory corruption, all of which can be systematically exploited by adversaries.

Over the past decade, DARPA investments have demonstrated that automated and semi-automated formal methods can substantially reduce these classes of vulnerabilities by exhaustively reasoning over software behavior. Contrary to common perception, such techniques are not monolithic or limited to niche expert use. Instead, they can be applied incrementally, with rigor increasing in proportion to mission criticality and acceptable cost, workforce maturity, and lifecycle constraints.

DARPA experience suggests that assurance rigor can be structured as graduated, ratcheting levels in which each successive level increases the degree of mechanization, automation, and formal reasoning applied to a software system. As an initial conceptual outline to stimulate discussion, such levels may include:

- Faithful, machine-readable recovery of software architecture and interfaces from code
- Mechanized descriptions of valid inputs and outputs at system interfaces
- Enforced memory safety guarantees across complete codebases
- Modeled and verified system properties aligned with model based system engineering (MBSE) artifacts
- Proven functional correctness and security under explicit threat models

These illustrative levels are not intended to be prescriptive, exhaustive, or fixed. Rather, they represent a notional progression to help quantify increasing degrees of formal methods application and to frame discussion around measurable assurance improvements. DARPA seeks community input to refine, restructure, validate, or replace this outline with a more technically grounded, operationally practical, and economically sustainable scale that can be adopted across development, sustainment, and modernization environments.

Through this RFI, DARPA seeks detailed recommendations and supporting evidence regarding how such graduated assurance levels can be technically defined, operationalized, validated,

economically justified, and transitioned into real-world programs and acquisition environments.

REQUESTED INFORMATION

This RFI seeks input on one or more of the following topics:

1. Graduated Levels of Software Assurance

Respondents should evaluate and refine DARPA's proposed conceptual outline for graduated levels of software assurance described in the Background section. Responses may restructure the levels, redefine their boundaries, introduce intermediate stages, propose alternative taxonomies, or identify additional technical dimensions necessary to produce a scalable and consumable framework for the performer base.

Responses should identify the technical mechanisms, assumptions, scalability constraints, tooling requirements, and measurable assurance properties associated with each proposed or revised level. Respondents should clearly explain how each level builds upon prior artifacts or tooling and how increasing rigor corresponds to elimination of defined vulnerability classes, measurable risk reduction, or improved system resilience.

Submissions should also address how assurance levels can be objectively characterized, validated, and compared across systems of differing size, architecture, language composition, and mission criticality. Where applicable, respondents are encouraged to provide quantitative metrics, empirical data, or case examples demonstrating how graduated rigor can be practically achieved and sustained.

2. Artifacts and Toolchains

Tools and workflows should be identified that pair machine-readable artifacts with automated or semi-automated analysis to ensure faithfulness between code, models, interfaces, and requirements. Respondents should describe how such artifacts are generated, validated, versioned, maintained, and reused across the system lifecycle while addressing mechanisms for ensuring consistency between artifacts and implementation, detecting drift, and preserving traceability across evolving requirements, models, and deployed code.

3. Incremental Adoption and Composition

Methods for gradually introducing higher assurance levels into legacy systems, mixed-language environments, or ongoing development programs without requiring full system redesign should be identified. This includes composability across assurance levels and partial adoption scenarios. Respondents should provide concrete integration strategies for incremental deployment in legacy or heterogeneous environments. Responses should address composability across differing assurance levels, partial adoption pathways, interoperability with existing tool chains, and techniques for isolating or encapsulating higher-assurance components within broader systems of mixed assurance.

4. Operational and Economic Considerations

Respondents should provide documented evidence, quantitative data, qualitative assessments, or case studies demonstrating how incremental assurance levels influence development velocity, sustainment costs, workforce skill requirements, defect density, vulnerability class elimination, and long-term system resilience.

Responses should characterize return on investment, productivity impacts, lifecycle cost implications, or other measurable operational outcomes, and should clearly distinguish between projected benefits and empirically validated results.

5. Transition and Acquisition Implications

Respondents should provide recommendations for how graduated assurance levels could be incorporated into government acquisition, sustainment, and supply chain processes.

Responses should describe how assurance levels may be structured within solicitations, statements of work, or performance specifications; how compliance, effectiveness, or maturity would be measured and verified; and how assurance requirements could be progressively ratcheted, enforced, and demonstrated across integrators, subcontractors, and suppliers over time.

Please see Appendix A for further details. Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, university-affiliated research centers, not-for-profit research institutions, and U.S. Government-sponsored labs.

SUBMISSION INSTRUCTIONS

Responses to this RFI should be submitted by **June 12, 2026, at 5:00 PM ET**.

Unclassified responses to this RFI should be submitted to ResilientSoftwareSystems@darpa.mil.

To the maximum extent possible, respondents should submit non-proprietary information. If proprietary information is submitted, it is the respondent's responsibility to clearly mark what is considered proprietary data. All proprietary information should be clearly labeled "Proprietary." Note, "confidential" is not a classification marking used to control the dissemination of U.S. Government National Security Information as dictated in Executive Order 13526 and should not be used to identify proprietary business information. DARPA will disclose submission contents only for the purpose of review by DARPA staff, other Government agencies, or DARPA Support Contractors.

Controlled Unclassified Information (CUI) Responses: For unclassified responses containing CUI, applicants will ensure personnel and information systems processing CUI security requirements are in place. If an unclassified submission contains CUI or the suspicion of such, as defined by Executive Order 13556 and 32 CFR Part 2002, the information must be appropriately and conspicuously marked CUI in accordance with DoDI 5200.48. Unclassified submissions containing CUI may be submitted via ResilientSoftwareSystems@darpa.mil. Entities generating and submitting CUI responses must do so on an information system authorized to process CUI information in accordance with National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 and Department of Defense Instruction (DoDI) 8582.01.

Classified Responses: At this time no classified responses are anticipated for this RFI. ResilientSoftwareSystems@darpa.mil. If a respondent would like to include classified materials in their response, please send an email to ResilientSoftwareSystems@darpa.mil stating intent to submit a classified response with an unclassified justification. If

appropriate, further coordination will occur to facilitate that classified response. NO CLASSIFIED INFORMATION SHOULD BE SENT TO ResilientSoftwareSystems@darpa.mil.

NOTE: DARPA may reach out to respondents to clarify and/or discuss technical information received against this RFI.

FORMAT AND CONTENT INSTRUCTIONS

DARPA is strongly encouraging all respondents to submit a single integrated response. All submissions must be submitted as an unprotected Microsoft Word document or PDF file. Submissions not meeting this requirement may be discarded.

All submissions must be written in English, and all pages shall be formatted for printing on 8-1/2 by 11-inch paper with 1-inch margins and font size no smaller than 12-point font. Font sizes of 8 or 10-point may be used for figures, tables, and charts. Submissions should be concise and must not exceed 10 pages; submissions in excess of the page limit may not be reviewed.

Page limit includes:	Page limit does NOT include:
Technical feedback	Cover Sheet
Technical figures, tables, charts	References (Optional)

See Appendix A for the RFI submission template, which describes content requirements.

DARPAConnect

Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare your ideas for high-impact conversations with DARPA program managers. Please visit [DARPAConnect.us](https://darpaconnect.us) to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss your specific ideas, questions, and paths to DARPA. You can use the contact form at [DARPAConnect.us](https://darpaconnect.us) or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

ADMINISTRATIVE

This RFI contains all information required to submit a response. No additional forms, kits, or other materials are needed. All administrative and technical questions should be directed to ResilientSoftwareSystems@darpa.mil. Please refer to the Special Notice number (DARPA-SN-26-71) in all correspondence.

This RFI does not constitute a formal solicitation for abstracts or proposals; any abstract or proposal received against this RFI will be disregarded. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. DARPA will not provide reimbursement for costs incurred in responding to this RFI. Respondents are advised that DARPA is under no obligation to acknowledge receipt of

the submission received, invite respondents to events, include tools identified in responses, or provide feedback to respondents with respect to any information submitted under this RFI.

**APPENDIX A
RFI SUBMISSION TEMPLATE
DARPA-SN-26-71**

COVER SHEET

Submission Title	
Submitter Organization	
Technical Point of Contact (POC)	Name: Address: Telephone: Email:
Administrative POC	Name: Address: Telephone: Email:
SAM.gov Unique Entity ID (UEI)¹	

¹ The SAM-generated UEI is the official identifier for doing business with the U.S. Government.

[Content requirements are stated in blue font and should be deleted prior to proposal submission. Responses to the RFI should cover at least one of the following questions.]

Respondents should provide information, technical approaches, analyses, and recommendations describing how graduated, incrementally adoptable levels of increasingly rigorous, mechanized, and automated software assurance can be defined, implemented, measured, and sustained across development, sustainment, and modernization activities. Responses should clearly articulate practical implementation strategies, measurable outcomes, and transition considerations associated with each proposed assurance level.

DARPA experience suggests that assurance rigor can be structured as graduated, ratcheting levels in which each successive level increases the degree of mechanization, automation, and formal reasoning applied to a software system. As an initial conceptual outline to stimulate discussion, such levels may include:

- Faithful, machine-readable recovery of software architecture and interfaces from code
- Mechanized descriptions of valid inputs and outputs at system interfaces
- Enforced memory safety guarantees across complete codebases
- Modeled and verified system properties aligned with MBSE artifacts
- Proven functional correctness and security under explicit threat models

These illustrative levels are not intended to be prescriptive, exhaustive, or fixed. Rather, they represent a notional progression to help quantify increasing degrees of formal methods application and to frame discussion around measurable assurance improvements. DARPA seeks community input to refine, restructure, validate, or replace this outline with a more technically grounded, operationally practical, and economically sustainable scale that can be adopted across development, sustainment, and modernization environments.